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(54) **AIR CONDITIONING SYSTEM AND
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(57) **ABSTRACT**

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An air conditioning system of an embodiment includes an indoor air handling unit controller and an outdoor air handling unit controller. The indoor air handling unit controller is configured to increase and decrease an operating capacity of the indoor air handling unit. The outdoor air handling unit controller is configured to increase and decrease an operating capacity of the outdoor air handling unit.

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2023/
043166, filed on Dec. 1, 2023.

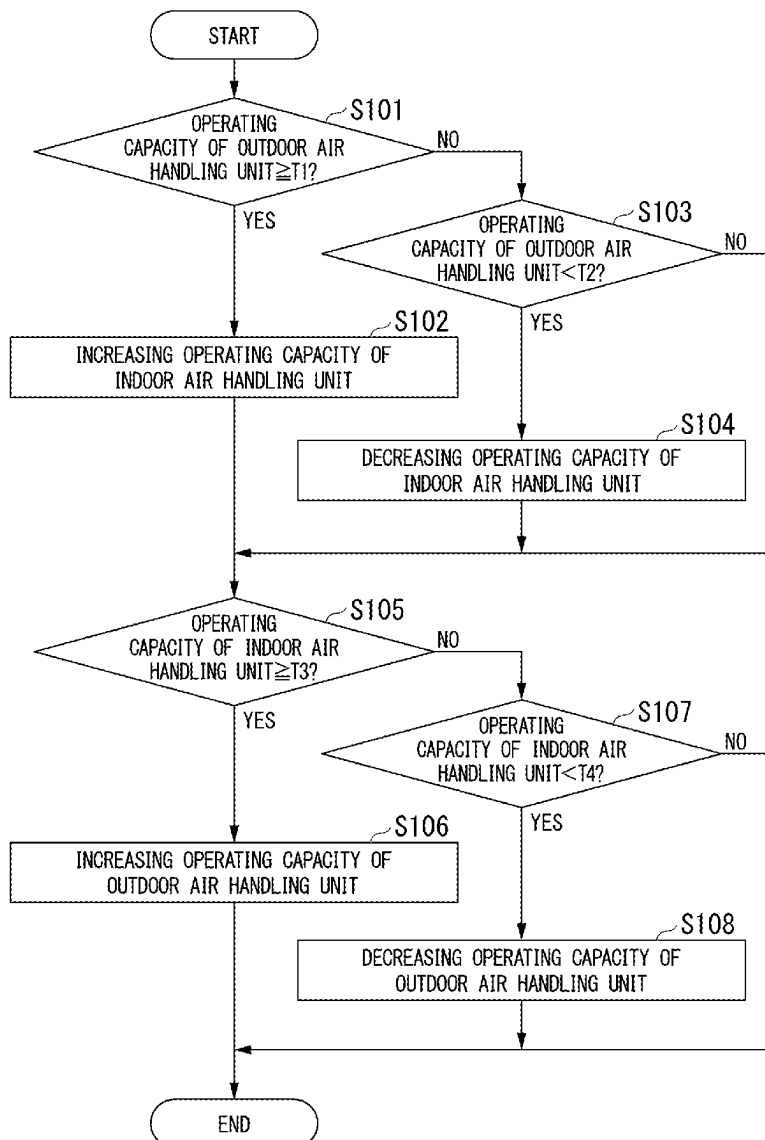


FIG. 2

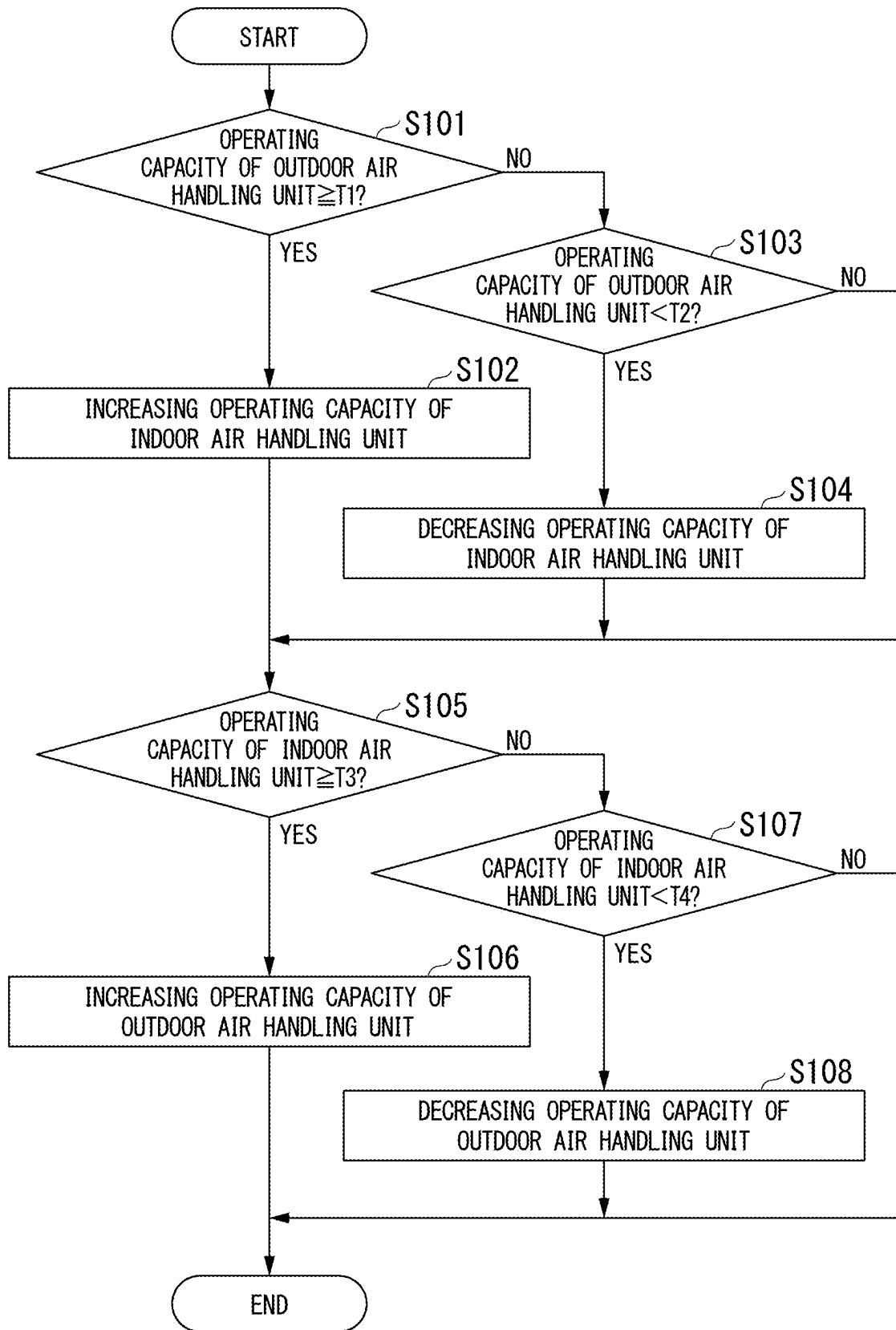
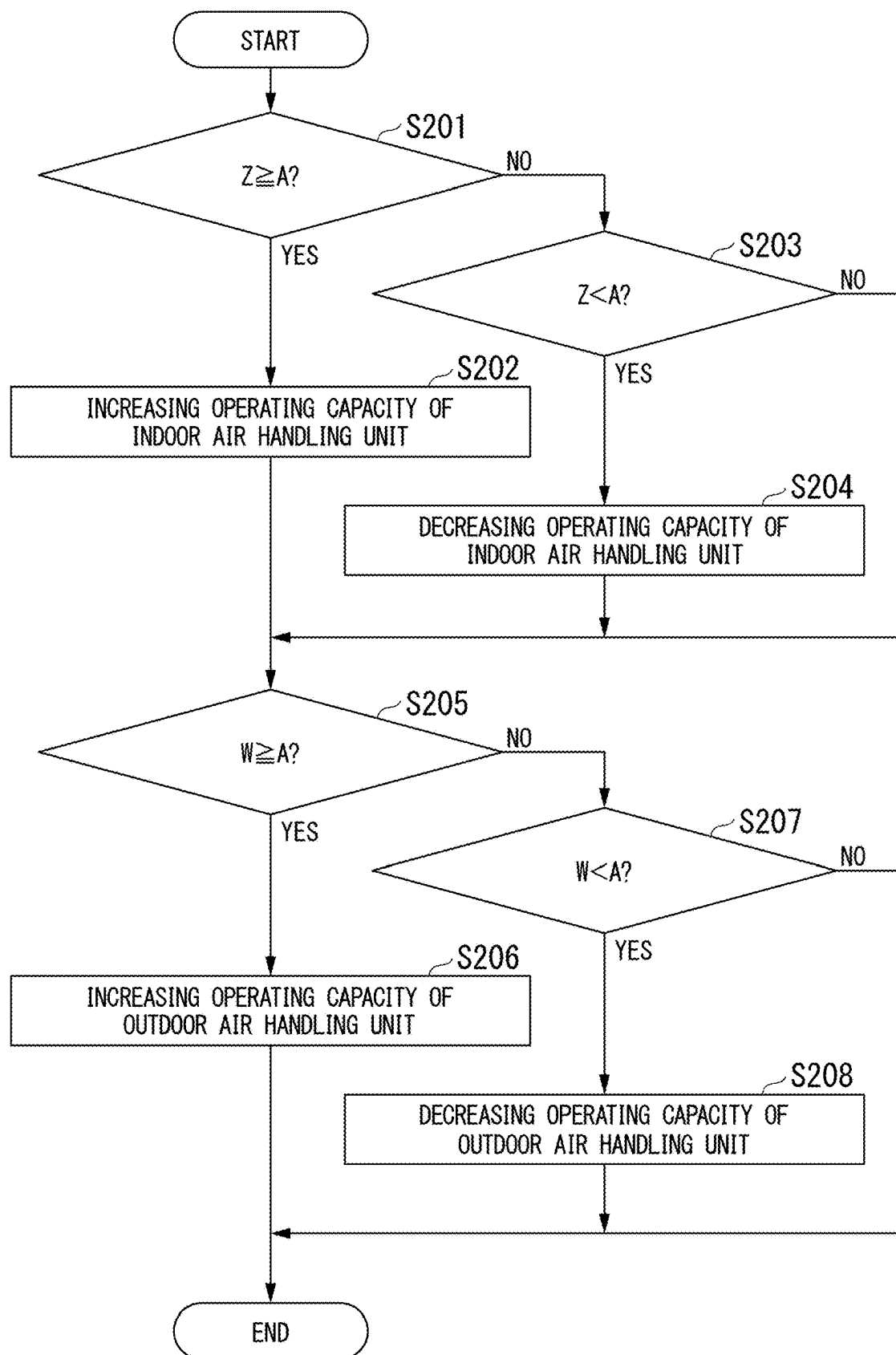


FIG. 3



AIR CONDITIONING SYSTEM AND CONTROL METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of International Application No. PCT/JP2023/043166, filed Dec. 1, 2023, which is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-192823, filed on Dec. 1, 2022; the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present invention relates to an air conditioning system and a control method. Priority is claimed on Japanese Patent Application No. 2022-192823 filed on Dec. 1, 2022, the contents of which are incorporated herein by reference.

BACKGROUND

[0003] In conventional technologies, coordinated control of an indoor air handling unit and outdoor air handling unit was aimed at reducing electric energy consumption, and was control of carrying out comparison of a coefficient of performance (COP) or comparison of power consumption (for example, Japanese Patent No. 6414354).

BRIEF DESCRIPTION OF DRAWINGS

[0004] FIG. 1 is a diagram showing a configuration example of an air conditioning system of an embodiment.

[0005] FIG. 2 is a flowchart showing a specific example of a processing flow of an air conditioning system in a first embodiment.

[0006] FIG. 3 is a flowchart showing a specific example of a processing flow of an air conditioning system in a second embodiment.

DETAILED DESCRIPTION

[0007] Hereinafter, an air conditioning system and a control method according to an embodiment will be described with reference to the drawings.

[0008] An air conditioning system of an embodiment includes an indoor air handling unit controller and an outdoor air handling unit controller. The indoor air handling unit controller is configured to: increase an operating capacity of an indoor air handling unit in a case in which an operating capacity of an outdoor air handling unit is greater than or equal to a first threshold value and the operating capacity of the indoor air handling unit is able to be increased; and decrease the operating capacity of the indoor air handling unit in a case in which the operating capacity of the outdoor air handling unit is less than a second threshold value that is lower than the first threshold value and the operating capacity of the indoor air handling unit can be decreased. The outdoor air handling unit controller is configured to: increase the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is greater than or equal to a third threshold value and the operating capacity of the outdoor air handling unit is able to be increased; and decrease the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is less than a fourth threshold value that is lower than

the third threshold value and the operating capacity of the outdoor air handling unit can be decreased.

[0009] FIG. 1 is a diagram showing a configuration example of an air conditioning system 17 of an embodiment. The air conditioning system 17 shown in FIG. 1 is a system that performs air conditioning for a plurality of spaces (in FIG. 1, two spaces 23 and 24 provided in a building 14). The air conditioning system 17 includes a controller 10, a room-dedicated outdoor unit 12, and a heat source unit 40. Also, the building 14 includes an air conditioner controller 18, two indoor air handling units 21, an exhaust duct 16, two variable air volume systems (VAVs) 20, and two remote controllers 22. The indoor air handling units 21, the VAVs 20, and the remote controllers 22 are each provided in the spaces 23 and 24.

[0010] The room-dedicated outdoor unit 12 adjusts a temperature of indoor air in the spaces 23 and 24. The remote controller 22 is used to set starting and stopping of the operation, an airflow, a temperature, and the like. The set contents are output to the controller 10. The controller 10 controls an operating state of an outdoor air handling unit 31 and the room-dedicated outdoor unit 12 according to the contents set by the remote controller 22, temperatures and humidities of outdoor air and indoor air, and the like.

[0011] The exhaust duct 16 is connected to an exhaust port (not shown in the drawings) provided in the spaces 23 and 24. The exhaust duct 16 is a duct for exhausting air in the spaces 23 and 24 to the outside via a building 15.

[0012] The building 15 includes a ventilation fan 26, an outdoor air handling unit controller 27, the outdoor air handling unit 31, and an air intake port 34. The ventilation fan 26 exhausts the air from the exhaust duct 16 described above. The outdoor air handling unit controller 27 controls the outdoor air handling unit 31. The air intake port 34 takes in outdoor air into the outdoor air handling unit 31.

[0013] The outdoor air handling unit 31 includes an outdoor air handling unit blower 29, a humidifier 30, and a water coil 32. The outdoor air handling unit 31 takes in outdoor air through an air intake duct 28 from the air intake port 34. The outdoor air handling unit 31 conditions (cools, heats, dehumidifies, or humidifies) the outdoor air taken in. The outdoor air handling unit 31 supplies the conditioned outdoor air from the VAV 20 to the spaces 23 and 24 via a supply air duct 25. The outdoor air handling unit 31 exhausts air from the spaces 23 and 24 to the outside of the building 13 through the exhaust duct 16 using the ventilation fan 26.

[0014] The water coil 32 is connected to the heat source unit 40. The water coil 32 is a heat exchanger that functions as a cooler for the outdoor air. The water coil 32 has a heat transfer tube and a heat transfer fin. In the water coil 32, heat exchange is performed between outdoor air passing around the heat transfer tube and the heat transfer fin, and a heat medium passing through the heat transfer tube. The humidifier 30 humidifies the outdoor air that has passed through the water coil 32. The outdoor air handling unit blower 29 is a blower that takes in outdoor air into the outdoor air handling unit 31 and sends it to the supply air duct 25.

[0015] Various types of sensors are provided in the outdoor air handling unit 31. Examples of sensors provided in the outdoor air handling unit 31 include an outdoor air temperature sensor and an outdoor air humidity sensor which detect a temperature and a humidity of outdoor air drawn into the outdoor air handling unit 31. Also, examples of sensors provided in the outdoor air handling unit 31

include a sensor that detects a temperature of the supply air sent to the supply air duct 25.

[0016] The supply air duct 25 obtains outdoor air by the air blown by the outdoor air handling unit blower 29. The supply air duct 25 is connected to the outdoor air handling unit 31 and the VAV 20. The outdoor air handling unit controller 27 is constituted by a calculation device, a storage device, various types of electrical components, and the like. The outdoor air handling unit controller 27 is electrically connected to the controller 10 and the remote controller 22 via a communication line. The outdoor air handling unit controller 27 controls an operating capacity of the outdoor air handling unit 31 according to settings made via the remote controller and a temperature of the outdoor air taken in.

[0017] The room-dedicated outdoor unit 12 generally includes a plurality of outdoor units and a plurality of (two in FIG. 1) indoor air handling units 21. The outdoor unit is connected to the indoor air handling unit 21 via a refrigerant communication pipe 19. Various types of sensors are provided in the outdoor unit and the indoor air handling unit 21. Examples of sensors provided in the outdoor unit include a suction pressure sensor that detects a pressure of the refrigerant drawn into the compressor, and a discharge pressure sensor that detects a pressure of the refrigerant discharged from the compressor. Examples of sensors provided in the indoor air handling unit 21 include sensors that detect a temperature, a humidity, and a concentration of carbon dioxide of the indoor air drawn into the indoor air handling unit 21.

[0018] The air conditioner controller 18 is an example of an indoor air handling unit controller. The air conditioner controller 18 is constituted by a calculation device, a storage device, various types of electrical components, and the like. The air conditioner controller 18 is connected to the room-dedicated outdoor unit 12, the indoor air handling unit 21, sensors of the indoor air handling unit 21, and the remote controller 22 via a communication line. The air conditioner controller 18 controls operating capacities of the room-dedicated outdoor unit 12 and the indoor air handling unit 21 according to settings made by the remote controller and an indoor temperature.

[0019] The controller 10 controls the entire air conditioning system 17. The controller 10 is constituted by a calculation device and a storage device.

[0020] Communication lines L1, L2, and L3 are communication lines for performing communication between the air conditioner controller 18 and the outdoor air handling unit controller 27. The air conditioner controller 18 communicates via the room-dedicated outdoor unit 12. Also, the outdoor air handling unit controller 27 communicates via the heat source unit 40. As a mode of performing communication between the air conditioner controller 18 and the outdoor air handling unit controller 27, there is a first communication mode that uses the communication line L1, and a second communication mode that uses the communication lines L2 and L3. The first communication mode is a mode in which communication is performed directly without going through the controller 10. In a case in which communication is performed in the first communication mode, the communication lines L2 and L3 are not necessary. The second communication mode is a mode in which the room-dedicated outdoor unit 12 and the controller 10 are connected by the communication line L2, and the heat source

unit 40 and the controller 10 are connected by the communication line L3. In this way, the second communication mode is a mode in which communication is performed via the controller 10. In a case in which communication is performed in the second communication mode, the communication line L1 is not necessary. Note that, as for a communication method, as long as it is sufficient that the air conditioner controller 18 and the outdoor air handling unit controller 27 can communicate with each other, and for example, a general-purpose protocol such as Modbus may be used.

[0021] Based on the above configuration, a first embodiment and a second embodiment will be described.

First Embodiment

[0022] In a first embodiment, the air conditioner controller 18 and the outdoor air handling unit controller 27 are capable of communicating with each other in the first communication mode. The air conditioner controller 18 is configured to be able to acquire an operating capacity of the outdoor air handling unit 31 through communication. The outdoor air handling unit controller 27 is configured to be able to acquire an operating capacity of the indoor air handling unit 21 via communication.

[0023] A first threshold value T1 for comparison with the operating capacity of the outdoor air handling unit 31 and a second threshold value T2 smaller than the first threshold value T1 are preset in the air conditioner controller 18. The first threshold value T1 and the second threshold value T2 are stored in the storage device of the air conditioner controller 18. The first threshold value T1 and the second threshold value T2 are set via, for example, the controller 10. In a case in which the operating capacity of the outdoor air handling unit 31 is greater than or equal to the first threshold value T1 and the operating capacity of the indoor air handling unit 21 can be increased, the air conditioner controller 18 increases the operating capacity of the indoor air handling unit 21. Also, in a case in which the operating capacity of the outdoor air handling unit 31 is less than the second threshold value T2 and the operating capacity of the indoor air handling unit 21 can be decreased, the air conditioner controller 18 decreases the operating capacity of the indoor air handling unit 21. The control for decreasing the operating capacity of the indoor air handling unit 21 also includes control for stopping the operation of the indoor air handling unit 21.

[0024] A third threshold value T3 for comparison with the operating capacity of the indoor air handling unit 21 and a fourth threshold value T4 smaller than the third threshold value T3 are preset in the outdoor air handling unit controller 27. The third threshold value T3 and the fourth threshold value T4 are stored in the storage device of the outdoor air handling unit controller 27. The third threshold value T3 and the fourth threshold value T4 are set via, for example, the controller 10. In a case in which the operating capacity of the indoor air handling unit 21 is greater than or equal to the third threshold value T3 and the operating capacity of the outdoor air handling unit 31 can be increased, the outdoor air handling unit controller 27 increases the operating capacity of the outdoor air handling unit 31. Also, in a case in which the operating capacity of the indoor air handling unit 21 is less than the fourth threshold value T4 and the operating capacity of the outdoor air handling unit 31 can be decreased, the outdoor air handling unit controller 27

decreases the operating capacity of the outdoor air handling unit 31. The control for decreasing the operating capacity of the outdoor air handling unit 31 also includes control for stopping the operation of the outdoor air handling unit 31.

[0025] FIG. 2 is a flowchart showing a processing flow of the air conditioning system 17 in the first embodiment. The air conditioner controller 18 determines whether the operating capacity of the outdoor air handling unit 31 is greater than or equal to the threshold value T1 (step S101). In a case in which the operating capacity of the outdoor air handling unit 31 is greater than or equal to the threshold value T1 (step S101: YES) and in a case in which the operating capacity of the indoor air handling unit 21 can be increased, the air conditioner controller 18 increases the operating capacity of the indoor air handling unit 21 (step S102), and then proceeds to step S105.

[0026] In a case in which the operating capacity of the outdoor air handling unit 31 is not greater than or equal to the threshold value T1 (step S101: NO), the air conditioner controller 18 determines whether the operating capacity of the outdoor air handling unit 31 is less than the threshold value T2 (step S103). In a case in which the operating capacity of the outdoor air handling unit 31 is less than the threshold value T2 (step S103: YES) and in a case in which the operating capacity of the indoor air handling unit 21 can be decreased, the air conditioner controller 18 decreases the operating capacity of the indoor air handling unit 21 (step S104), and then proceeds to step S105. In a case in which the operating capacity of the outdoor air handling unit 31 is not less than the threshold value T2 (step S103: NO), the processing proceeds to step S105.

[0027] The outdoor air handling unit controller 27 determines whether the operating capacity of the indoor air handling unit 21 is greater than or equal to the threshold value T3 (step S105). In a case in which the operating capacity of the indoor air handling unit 21 is greater than or equal to the threshold value T3 (step S105: YES) and in a case in which the operating capacity of the outdoor air handling unit 31 can be increased, the outdoor air handling unit controller 27 increases the operating capacity of the outdoor air handling unit 31 (step S106) and ends the processing.

[0028] In a case in which the operating capacity of the indoor air handling unit 21 is not greater than or equal to the threshold value T3 (step S105: NO), the outdoor air handling unit controller 27 determines whether the operating capacity of the indoor air handling unit 21 is less than the threshold value T4 (step S107). In a case in which the operating capacity of the indoor air handling unit 21 is less than the threshold value T4 (step S107: YES) and in a case in which the operating capacity of the outdoor air handling unit 31 can be decreased, the outdoor air handling unit controller 27 decreases the operating capacity of the outdoor air handling unit 31 (step S108) and ends the processing. In a case in which the operating capacity of the indoor air handling unit 21 is not less than the threshold value T4 (step S107: NO), the processing ends.

[0029] The processing shown in FIG. 2 may be configured as loop processing that does not end at step S106 or step S108, but returns to step S101 after step S106 or step S108 is completed. Alternatively, this processing may be repeatedly performed at regular intervals (for example, every one

minute). This processing may also be performed in response to an instruction from an operator of the air conditioning system 17.

[0030] A flow of the basic processing in the first embodiment is as described above. As shown in the flowchart of FIG. 2, a purpose of the processing in the present embodiment is to suppress a load becoming biased toward either the outdoor air handling unit 31 or the indoor air handling unit 21. In a case in which it is possible to suppress a load becoming biased toward either the outdoor air handling unit 31 or the indoor air handling unit 21, processing may be added to or modified in the processing shown in the flowchart of FIG. 2.

[0031] Specifically, a processing example in a case in which the threshold values T1 and T3 are set to an operating capacity of 80%, and the threshold values T2 and T4 are set to an operating capacity of 70% will be described. First, in a case in which the operating capacity of the indoor air handling unit 21 reaches 80% or more, if set dehumidification performance is obtained, the air conditioner controller 18 increases an airflow of the indoor air handling unit 21, and the outdoor air handling unit controller 27 increases the operating capacity of the outdoor air handling unit. In a case in which the operating capacity of the indoor air handling unit 21 is less than the threshold value T1 and greater than or equal to the threshold value T3, the outdoor air handling unit controller 27 does not perform airflow correction control, but instead controls the operating capacity to be equalized with that of the indoor air handling unit 21. In a case in which the operating capacity of the outdoor air handling unit 31 is less than 70%, the air conditioner controller 18 gradually reduces the airflow correction and shifts to normal control. Here, the normal control refers to control of the room-dedicated outdoor unit 12 and the indoor air handling unit 21 that is performed according to settings made by the remote controller or an indoor temperature.

[0032] Note that, for example, in a case in which the operating capacity of the indoor air handling unit 21 reaches 80% or more and in a case in which the operating capacity of the outdoor air handling unit 31 is assumed to be 100%, the operating capacity of the outdoor air handling unit 31 cannot be increased. However, in a case in which the ventilation airflow is satisfactory, the outdoor air handling unit controller 27 reduces the load on the outdoor air handling unit 31 (reduces the airflow, changes the set temperature of the heat source unit, or reduces a water flow rate). Therefore, since the load on the outdoor air handling unit 31 can be reduced while the load on the indoor air handling unit 21 increases, it is possible to suppress the load becoming biased toward either the outdoor air handling unit 31 or the indoor air handling unit 21.

[0033] In a case in which controlled in this way, it is possible to suppress a load becoming biased toward either the outdoor air handling unit 31 or the indoor air handling unit 21. Therefore, since it is possible to avoid situations in which components of equipment operating under a high load are more likely to fail or a maintenance cycle of the equipment operating under a high load becomes shorter, running costs of the air conditioning system can be reduced.

Second Embodiment

[0034] In the second embodiment, the air conditioner controller 18 and the outdoor air handling unit controller 27 can communicate with each other in the first communication

mode or the second communication mode. The air conditioner controller **18** is configured to be able to acquire an operating capacity of the outdoor air handling unit **31** via communication. The outdoor air handling unit controller **27** is configured to be able to acquire an operating capacity of the indoor air handling unit **21** via communication.

[0035] A predetermined value $A (>0)$ for comparing a difference between the operating capacity of the outdoor air handling unit **31** and the operating capacity of the indoor air handling unit **21** is preset in the air conditioner controller **18**. The predetermined value A is stored in the storage device of the air conditioner controller **18**. Similarly, the predetermined value A is preset in the outdoor air handling unit controller **27**. The predetermined value A is stored in the storage device of the outdoor air handling unit controller **27**. The predetermined value A is set via, for example, the controller **10**. In a case in which a difference between an operating capacity X of the outdoor air handling unit **31** and an operating capacity Y of the indoor air handling unit **21** becomes greater than or equal to the predetermined value A ($X-Y>A$) and the operating capacity of the indoor air handling unit **21** can be increased, the air conditioner controller **18** increases the operating capacity of the indoor air handling unit **21**. Also, in a case in which the difference between the operating capacity of the outdoor air handling unit **31** and the operating capacity of the indoor air handling unit **21** becomes less than the predetermined value ($X-Y<A$) and the operating capacity of the indoor air handling unit **21** can be decreased, the air conditioner controller **18** decreases the operating capacity of the indoor air handling unit **21**. The control for decreasing the operating capacity of the indoor air handling unit **21** also includes control for stopping the operation of the indoor air handling unit **21**.

[0036] In a case in which a difference between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** becomes greater than or equal to the predetermined value A ($Y-X>A$) and the operating capacity of the outdoor air handling unit **31** can be increased, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31**. Also, in a case in which the difference between the operating capacity of the indoor air handling unit **21** and the operating capacity of the outdoor air handling unit **31** becomes less than the predetermined value ($Y-X<A$) and the operating capacity of the outdoor air handling unit **31** can be decreased, the outdoor air handling unit controller **27** decreases the operating capacity of the outdoor air handling unit **31**. The control for decreasing the operating capacity of the outdoor air handling unit **31** also includes control for stopping the operation of the outdoor air handling unit **31**.

[0037] FIG. 3 is a flowchart showing a processing flow of the air conditioning system **17** in the second embodiment. In the flowchart of FIG. 3, a difference ($X-Y$) between an operating capacity X of the outdoor air handling unit **31** and an operating capacity Y of the indoor air handling unit **21** is defined as Z . A difference ($Y-X$) between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is defined as W .

[0038] The air conditioner controller **18** determines whether the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** is greater than

or equal to the predetermined value A (step **S201**). In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** is greater than or equal to the predetermined value A (step **S201**: YES) and in a case in which the operating capacity of the indoor air handling unit **21** can be increased, the air conditioner controller **18** increases the operating capacity of the indoor air handling unit **21** (step **S202**), and then proceeds to step **S205**.

[0039] In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** is not greater than or equal to the predetermined value A (step **S201**: NO), the air conditioner controller **18** determines whether the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step **S203**). In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step **S203**: YES), and in a case in which the operating capacity of the indoor air handling unit **21** can be decreased, the air conditioner controller **18** decreases the operating capacity of the indoor air handling unit **21** (step **S204**), and then proceeds to step **S205**. In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** has not become less than the predetermined value A from a state greater than or equal to the predetermined value A (step **S203**: NO), the processing proceeds to step **S205**.

[0040] The outdoor air handling unit controller **27** determines whether a difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is greater than or equal to the predetermined value A (step **S205**). In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is greater than or equal to the predetermined value A (step **S205**: YES) and in a case in which the operating capacity of the outdoor air handling unit **31** can be increased, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31** (step **S206**) and ends the processing.

[0041] In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is not greater than or equal to the predetermined value A (step **S205**: NO), the outdoor air handling unit controller **27** determines whether the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step **S207**). In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step **S207**: YES), and in a case in which the operating capacity

of the outdoor air handling unit **31** can be decreased, the outdoor air handling unit controller **27** decreases the operating capacity of the outdoor air handling unit **31** (step **S208**) and ends the processing. In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** has not become less than the predetermined value A from a state greater than or equal to the predetermined value A (step **S207**: NO), the processing ends.

[0042] The above-described processing shown in FIG. **3** may be performed at regular intervals (for example, every one minute) or in response to an instruction from an operator of the air conditioning system **17**.

[0043] A flow of the basic processing in the second embodiment is as described above.

[0044] As shown in the flowchart of FIG. **3**, a purpose of the processing in the present embodiment is to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. In a case in which it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**, processing may be added to or modified in the processing shown in the flowchart of FIG. **3**.

[0045] Specifically, a processing example in a case in which the predetermined value A is set to 20% will be described. In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is 20% or more, this is assumed to be an operating capacity at which dehumidification performance of the outdoor air handling unit **31** can be obtained. In this case, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31** by increasing an airflow of the outdoor air handling unit **31** or lowering a blowing temperature. As a result, since the load is equalized, it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** becomes less than 20% from a state of 20% or more, the air conditioner controller **18** stops the operation of the indoor air handling unit **21**, and the outdoor air handling unit controller **27** performs normal control using only the outdoor air handling unit **31**. The normal control here refers to control by the outdoor air handling unit controller **27** that is performed according to settings made by the remote controller or a temperature of outdoor air taken in. As equipment having a higher load between the outdoor air handling unit **31** or the indoor air handling unit **21** is stopped in this way, an operation time of the equipment with a higher load is reduced.

[0046] In this way, it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. Therefore, since it is possible to avoid situations in which components of equipment operating under a high load are more likely to fail or a maintenance cycle of the equipment operating under a high load becomes shorter, running costs of the air conditioning system can be reduced.

[0047] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may

be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

What is claimed is:

1. An air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising:
 - an indoor air handling unit controller configured to
 - increase an operating capacity of the indoor air handling unit in a case in which an operating capacity of the outdoor air handling unit is greater than or equal to a first threshold value and the operating capacity of the indoor air handling unit is able to be increased, and
 - decrease the operating capacity of the indoor air handling unit in a case in which the operating capacity of the outdoor air handling unit is less than a second threshold value that is lower than the first threshold value and the operating capacity of the indoor air handling unit is able to be decreased; and
 - an outdoor air handling unit controller configured to
 - increase the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is greater than or equal to a third threshold value and the operating capacity of the outdoor air handling unit is able to be increased, and
 - decrease the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is less than a fourth threshold value that is lower than the third threshold value and the operating capacity of the outdoor air handling unit is able to be decreased.
2. An air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising:
 - an indoor air handling unit controller configured to
 - increase an operating capacity of the indoor air handling unit in a case in which a difference between an operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the indoor air handling unit is able to be increased, and
 - decrease the operating capacity of the indoor air handling unit in a case in which the difference between the operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes less than the predetermined value and the operating capacity of the indoor air handling unit is able to be decreased; and
 - an outdoor air handling unit controller configured to
 - increase the operating capacity of the outdoor air handling unit in a case in which a difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the outdoor air handling unit is able to be increased, and
 - decrease the operating capacity of the outdoor air handling unit in a case in which the difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes less than the predetermined value and the operating capacity of the outdoor air handling unit is able to be decreased.

handling unit and the operating capacity of the outdoor air handling unit becomes less than the predetermined value and the operating capacity of the outdoor air handling unit is able to be decreased.

3. A control method of an air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising:

increasing an operating capacity of the indoor air handling unit in a case in which an operating capacity of the outdoor air handling unit is greater than or equal to a first threshold value and the operating capacity of the indoor air handling unit is able to be increased, and decreasing the operating capacity of the indoor air handling unit in a case in which the operating capacity of the outdoor air handling unit is less than a second threshold value that is lower than the first threshold value and the operating capacity of the indoor air handling unit is able to be decreased; and

increasing the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is greater than or equal to a third threshold value and the operating capacity of the outdoor air handling unit is able to be increased, and decreasing the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is less than a fourth threshold value that is lower than the third threshold value and the operating capacity of the outdoor air handling unit is able to be decreased.

4. A control method of an air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising:

increasing an operating capacity of the indoor air handling unit in a case in which a difference between an operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the indoor air handling unit is able to be increased, and decreasing the operating capacity of the indoor air handling unit in a case in which the difference between the operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes less than the predetermined value and the operating capacity of the indoor air handling unit is able to be decreased; and

increasing the operating capacity of the outdoor air handling unit in a case in which a difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the outdoor air handling unit is able to be increased, and decreasing the operating capacity of the outdoor air handling unit in a case in which the difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes less than the predetermined value and the operating capacity of the outdoor air handling unit is able to be decreased.

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